

LeafFit: Plant-Asset Creation from 3D Gaussian Splatting

Chang Luo Umetani Nobuyuki

The University of Tokyo



Converting 3DGS-captured Plant to Asset-ready Mesh



Input: 3DGS



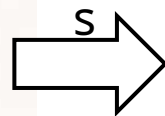
**Output:
Mesh**

Implicit method's output are heavy



3DGS Captured Plant

Implicit
Method



#Verts
1,856,309
#Faces
3,737,746

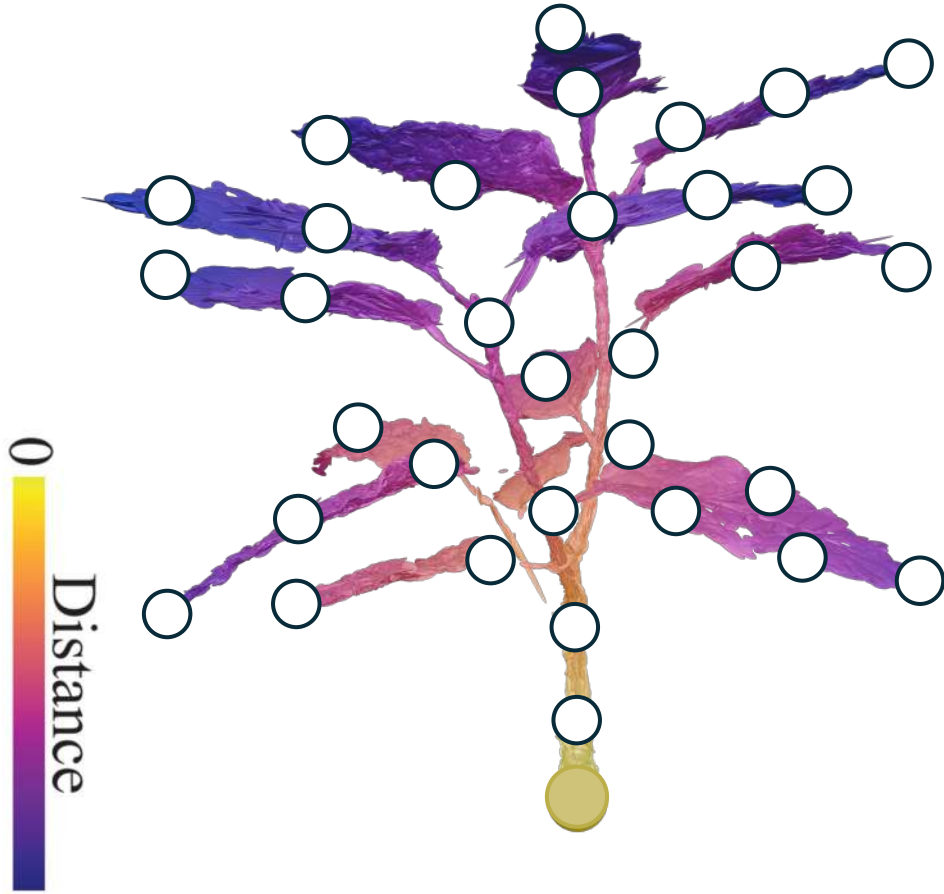
≈ 76 MB

Output Mesh

Estimate Geodesic Distance Fields

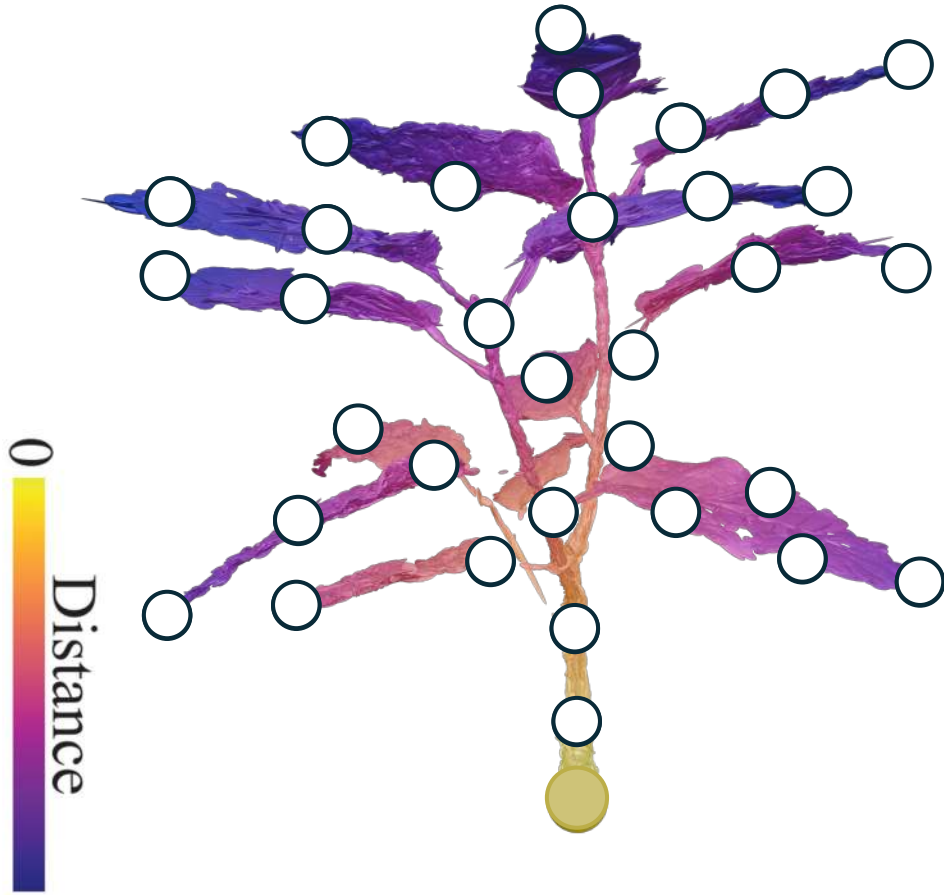


Down-sample



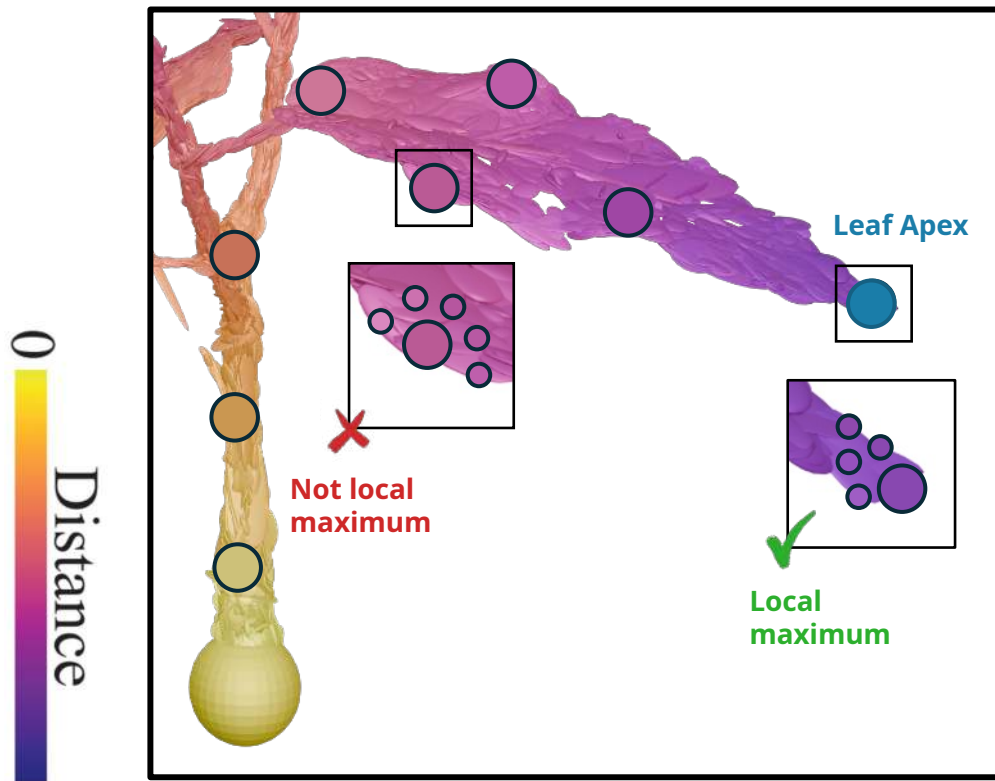
- Get sparse points ○ by down-sampling

Measure distance to root



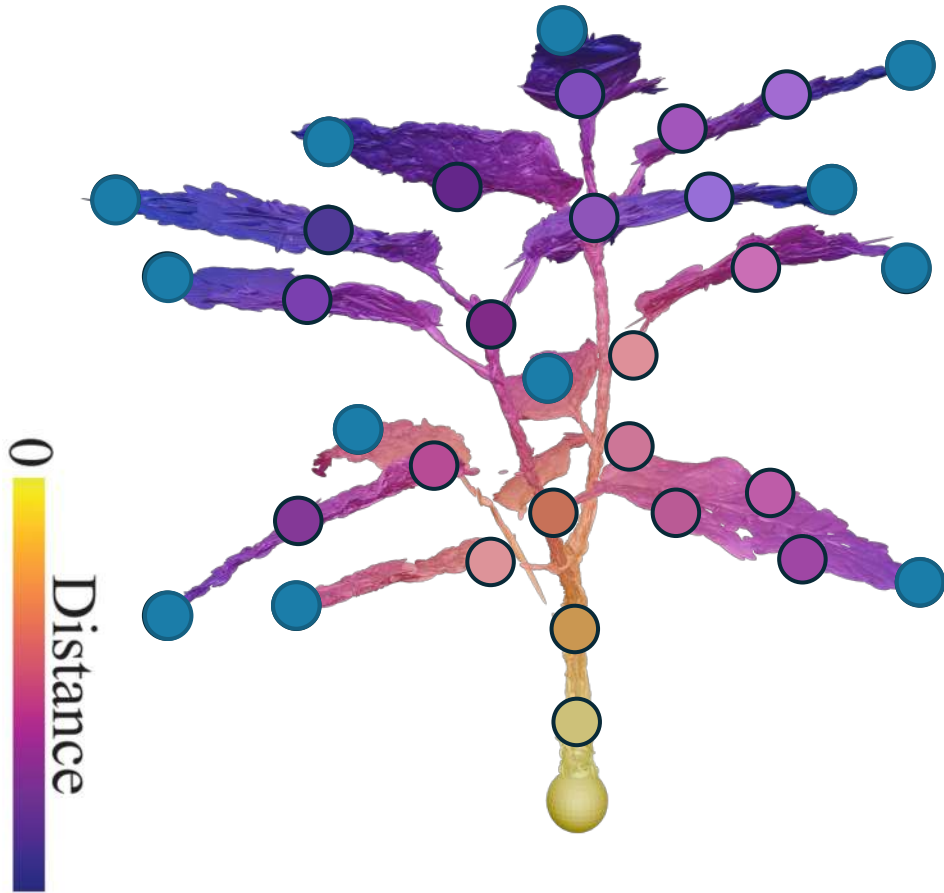
- Get sparse points $\bigcirc$ by down-sampling
- Get the geodesic distance to **root** $\bullet$

Determine leaf apexes



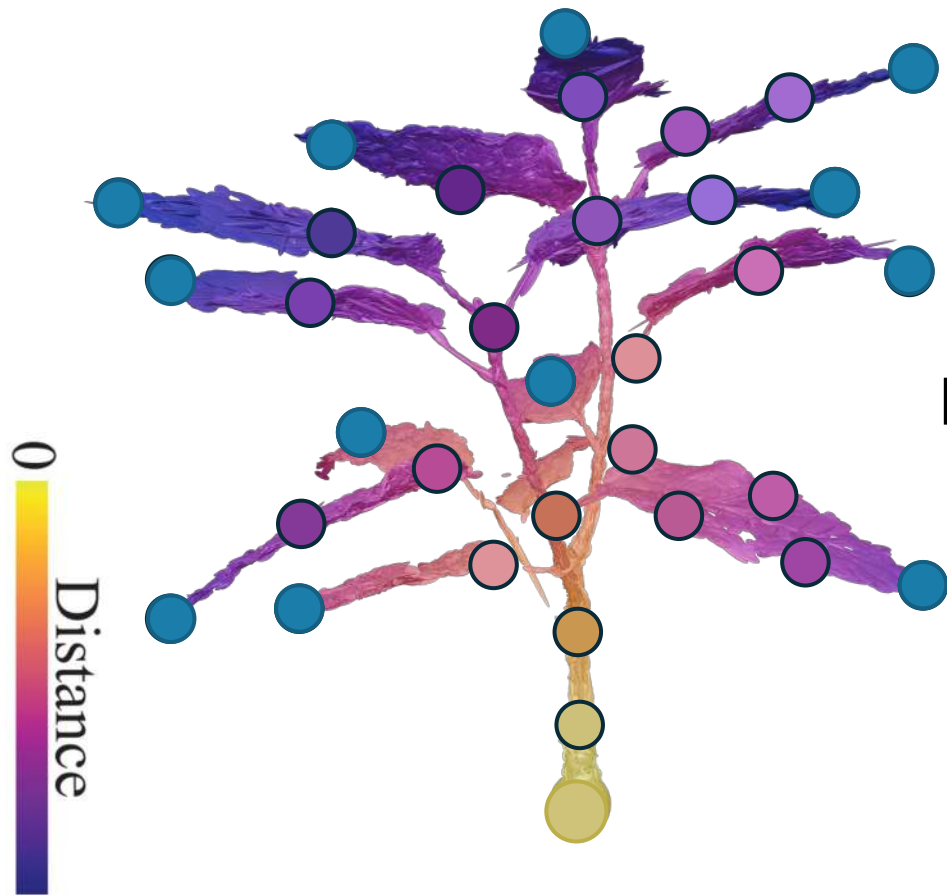
- Get sparse points ○ by down-sampling
- Get the geodesic distance to **root** ●
- Check neighbors to find leaf apices ●

Determine leaf apexes



- Get sparse points ○ by down-sampling
- Get the geodesic distance to **root** ●
- Check neighbors to find **leaf apices** ●

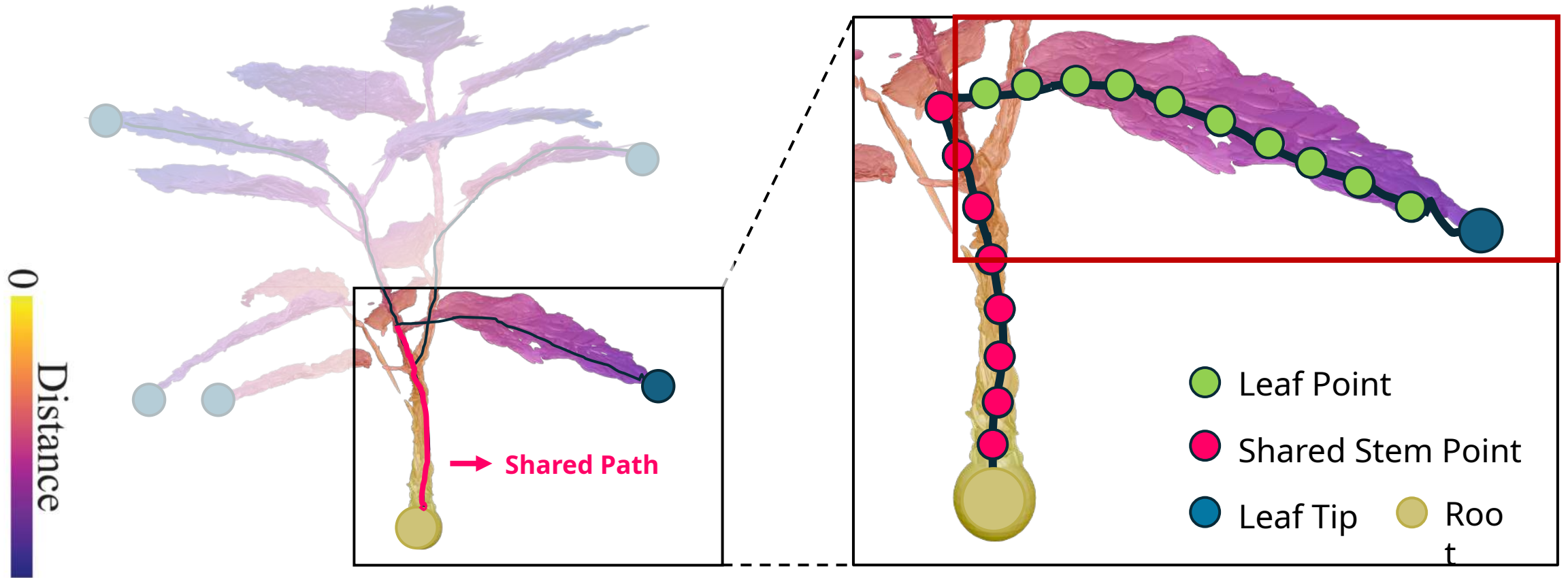
Get Apex-Root pathes



- Get sparse points ○ by down-sampling
- Get the geodesic distance to **root**
- Check neighbors to find **leaf**
- Get the paths from **apexes** to **root**

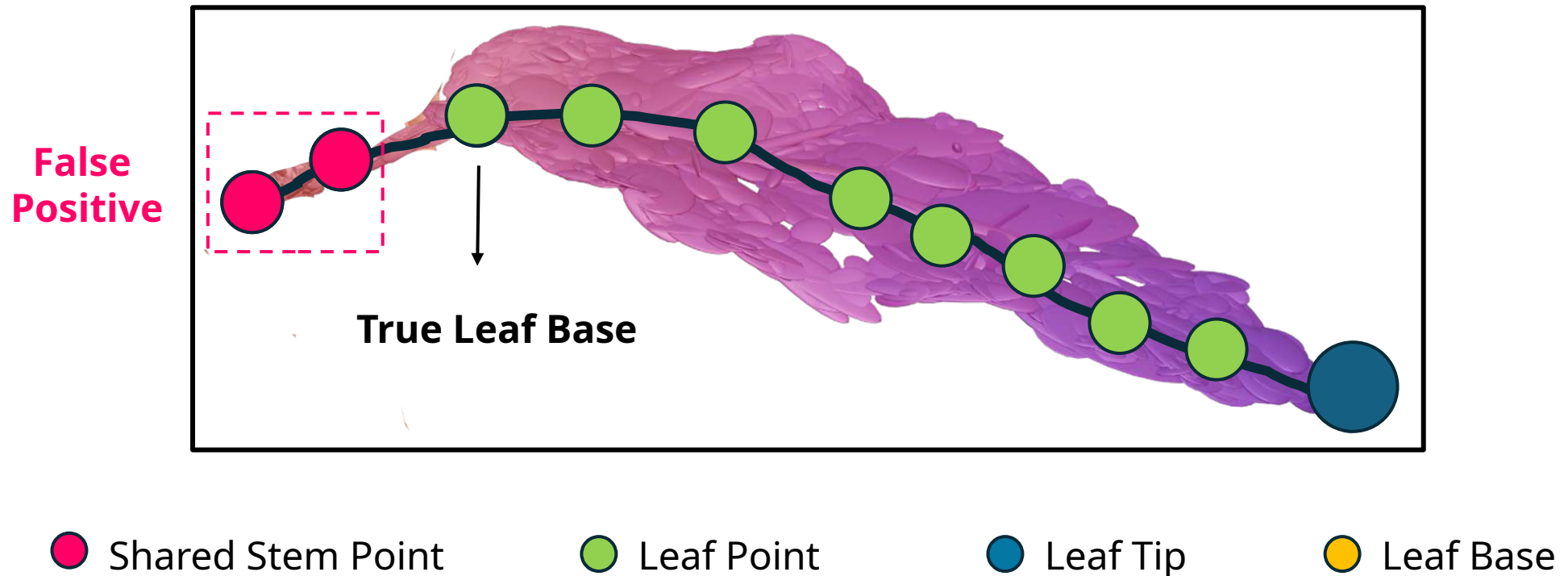


Leaf-stem Segmentation



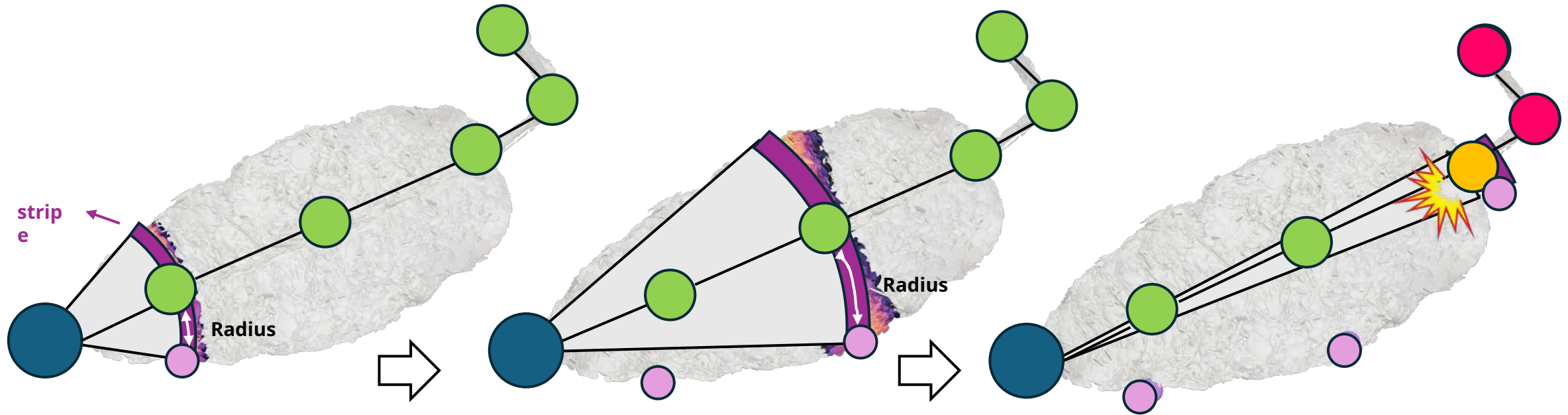
Leaf-stem Segmentation

Some points are misclassified as **leaf points**.



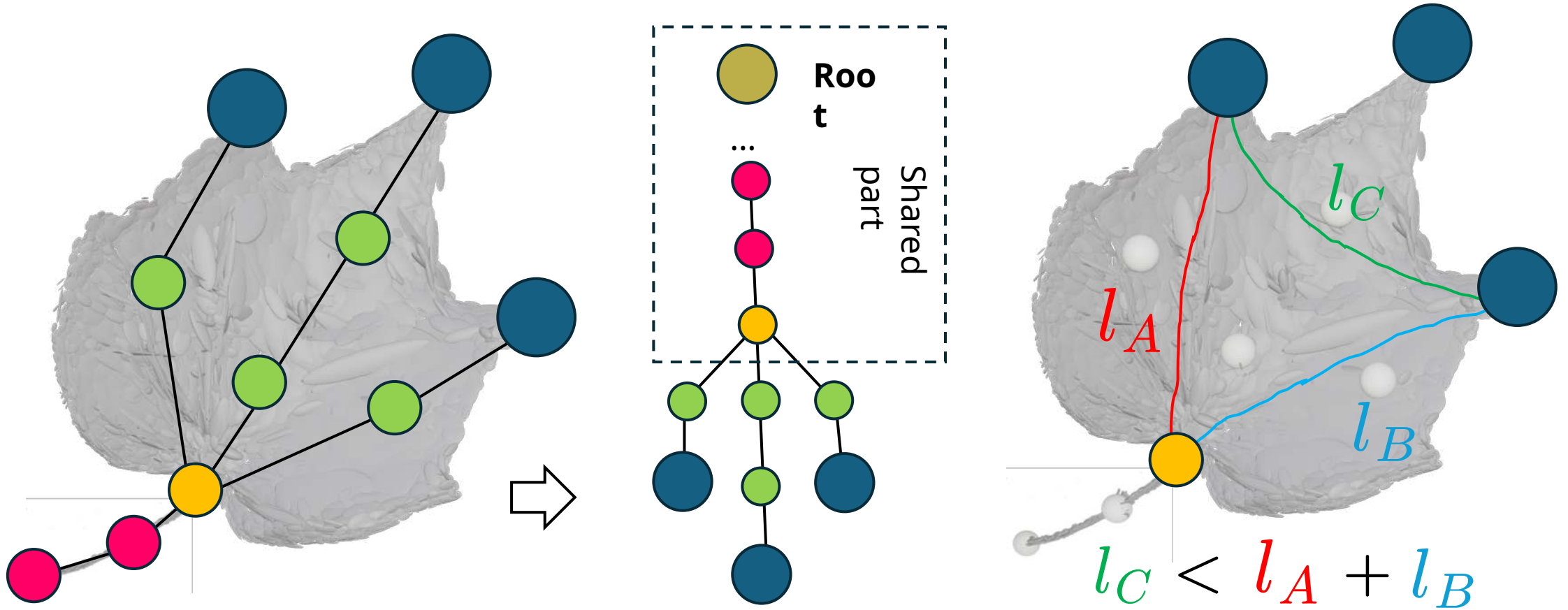
Find real leaf base

Radius is computed farthest distance from center (leaf points).
Leaf base has the smallest distance.



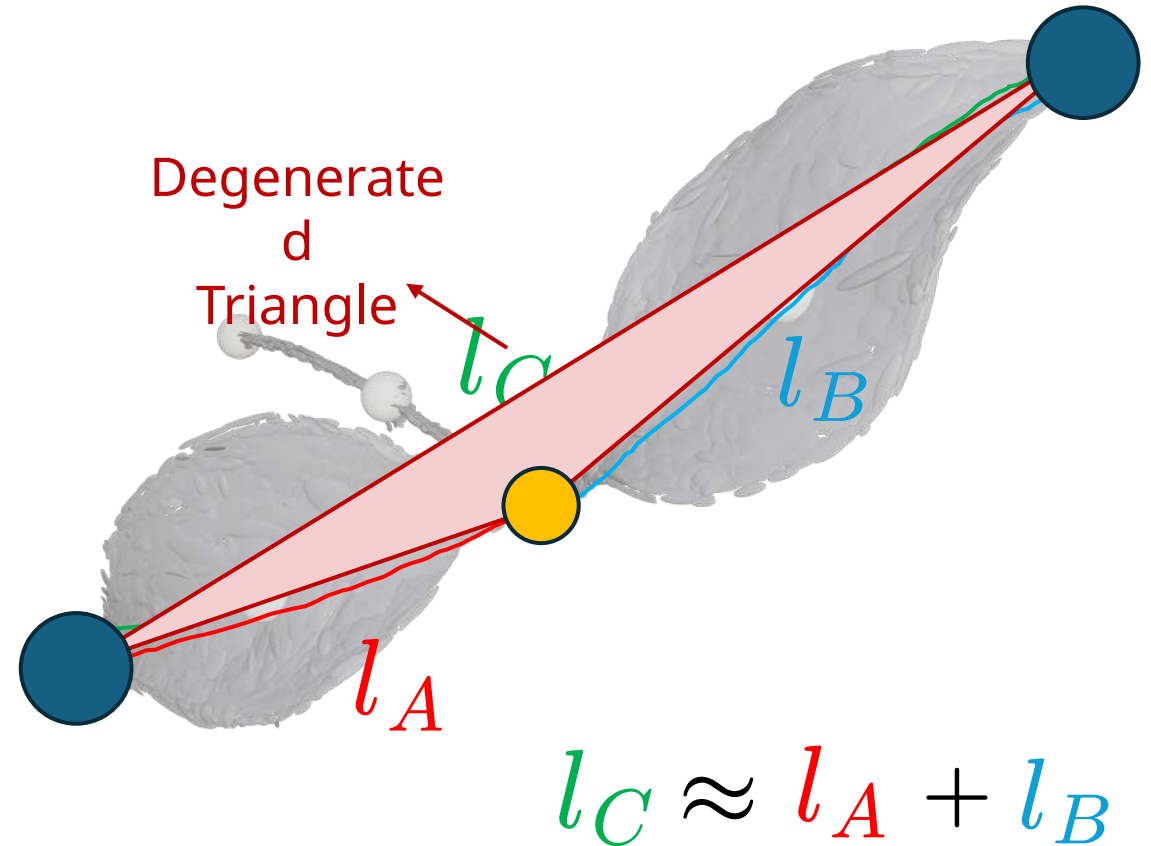
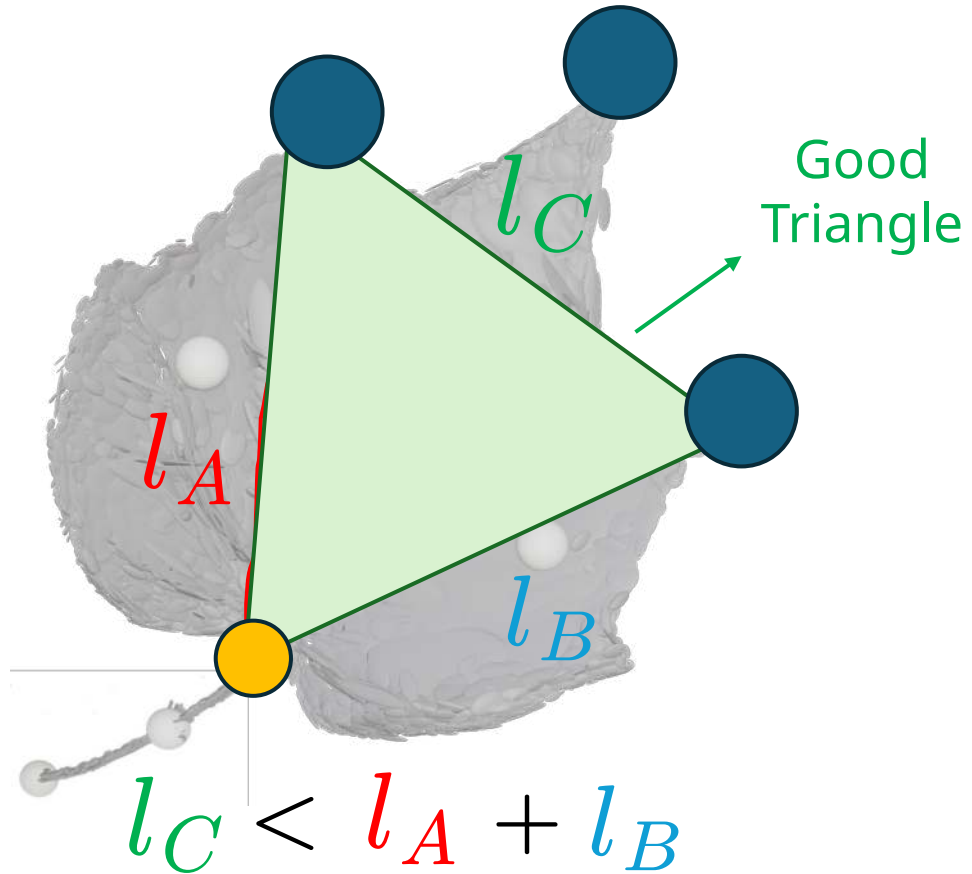
Tip Grouping for complicated leaves

- We group **tips** to where triangle inequality is unsatisfied.



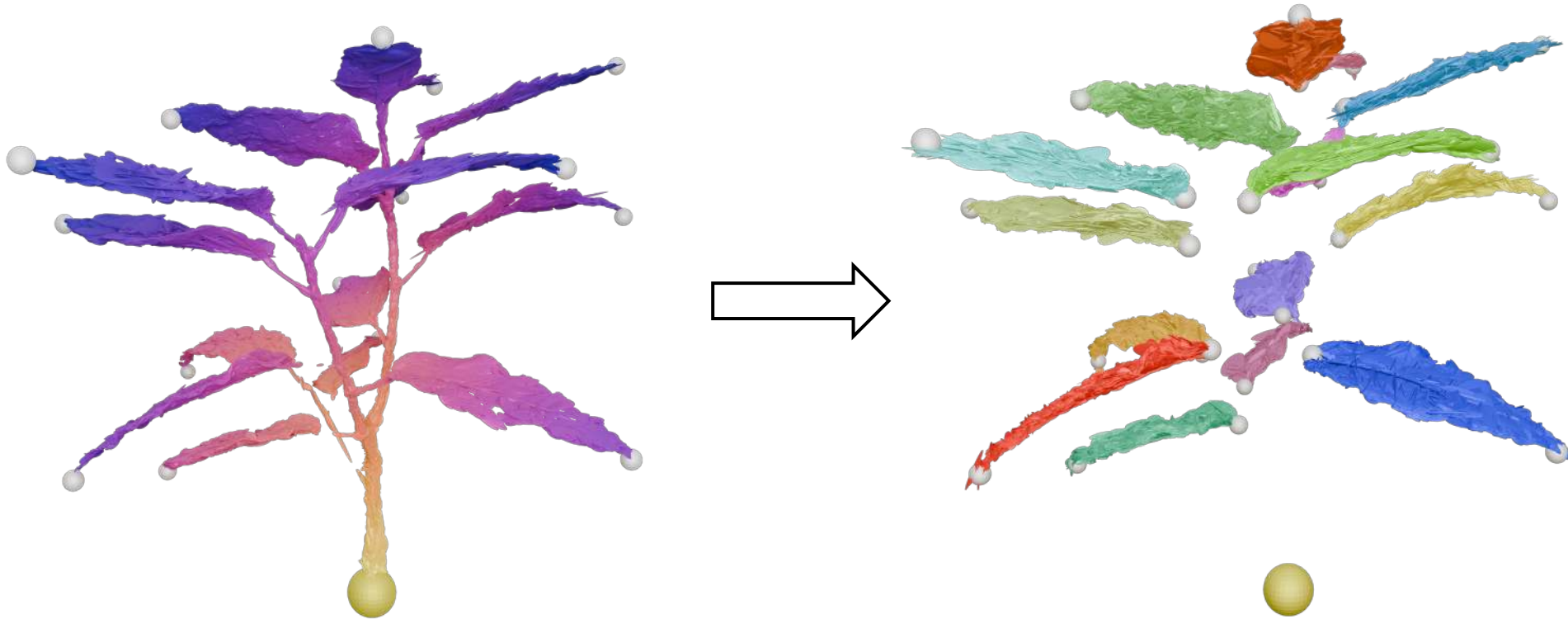
Tip Grouping for complicated leaves

Inequality of Triangle is used to group leaf tips.



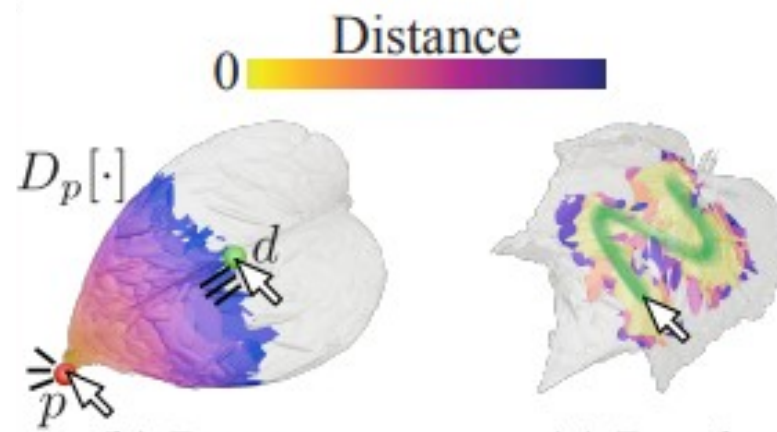
Automatic Segmentation

Leaves are well segmented now.



Manual fallback is also possible.

x8

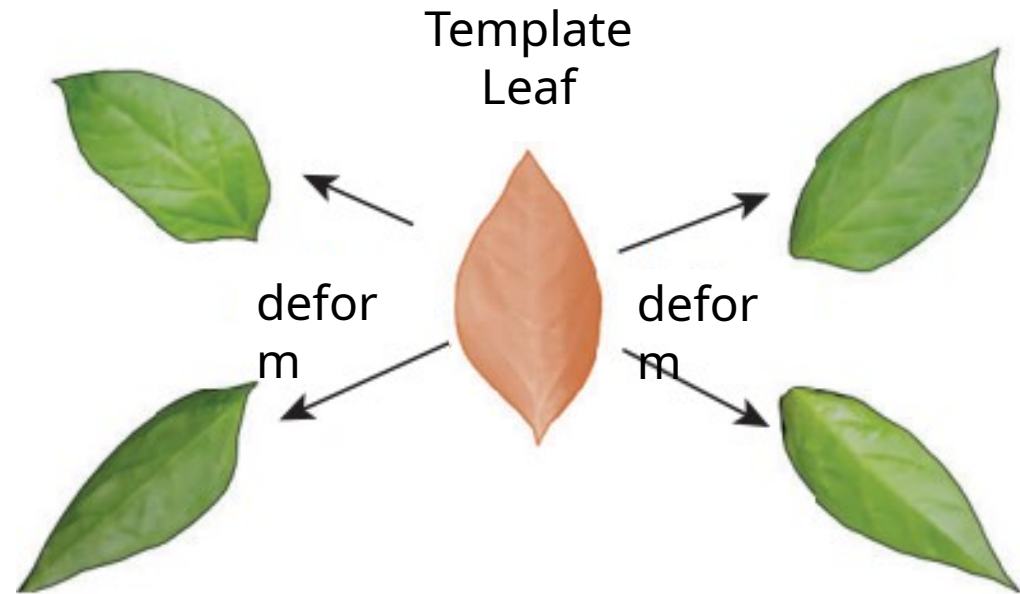


Dra
g

Brush

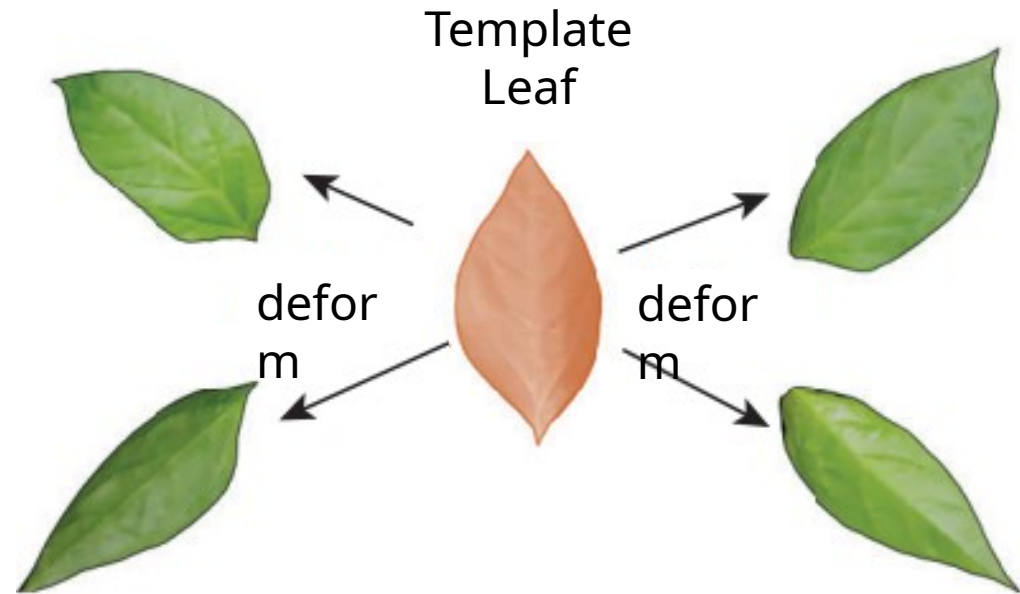
Compression using Similarity

Leaves are similar, take this inherent similarity to compress.



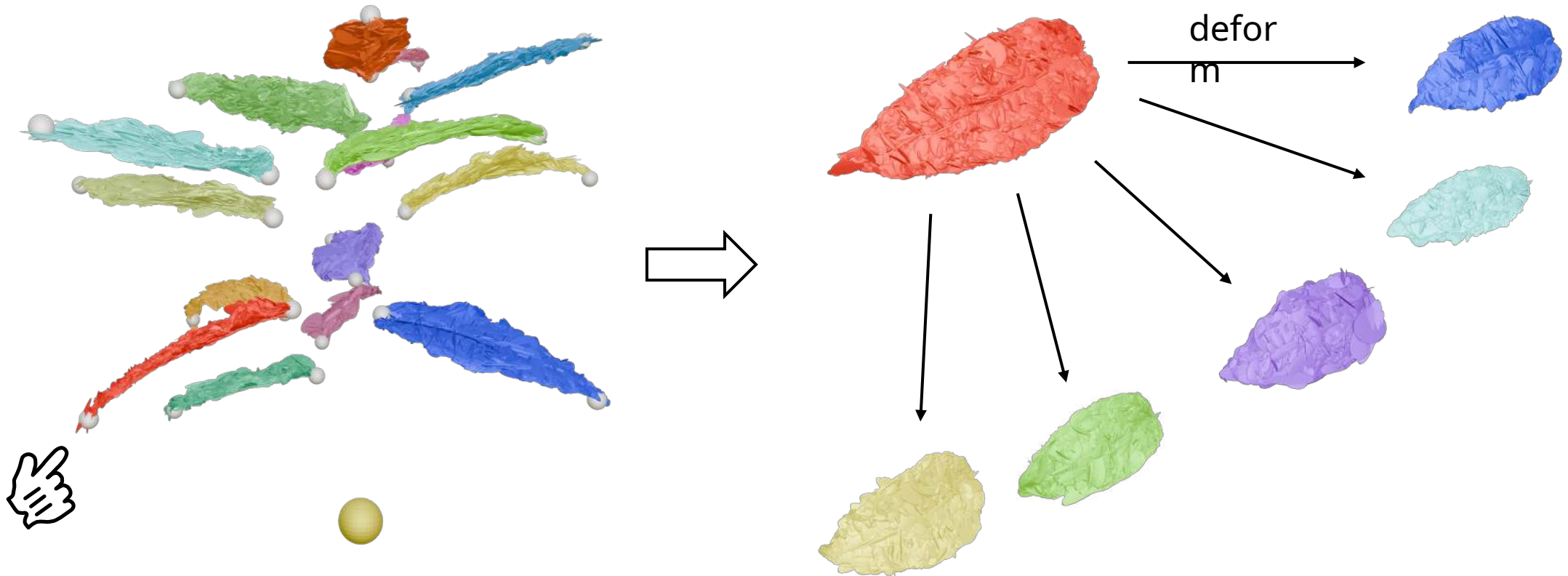
Compression using Similarity

Leaves are similar, take this inherent similarity to compress.



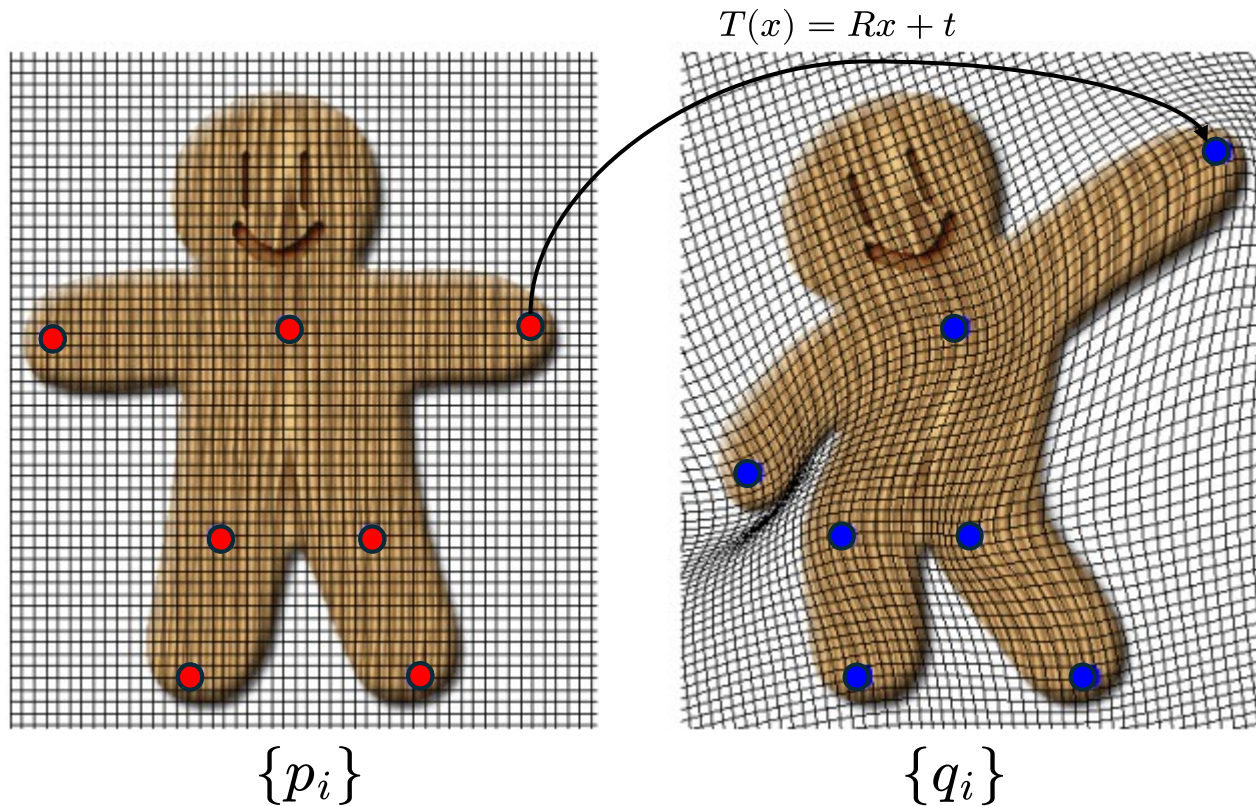
Similarity

Leaves are similar, take this inherent similarity to compress.



Moving Least Squares

Moving Least Squares [Schaefer+06] optimizes local transformation.



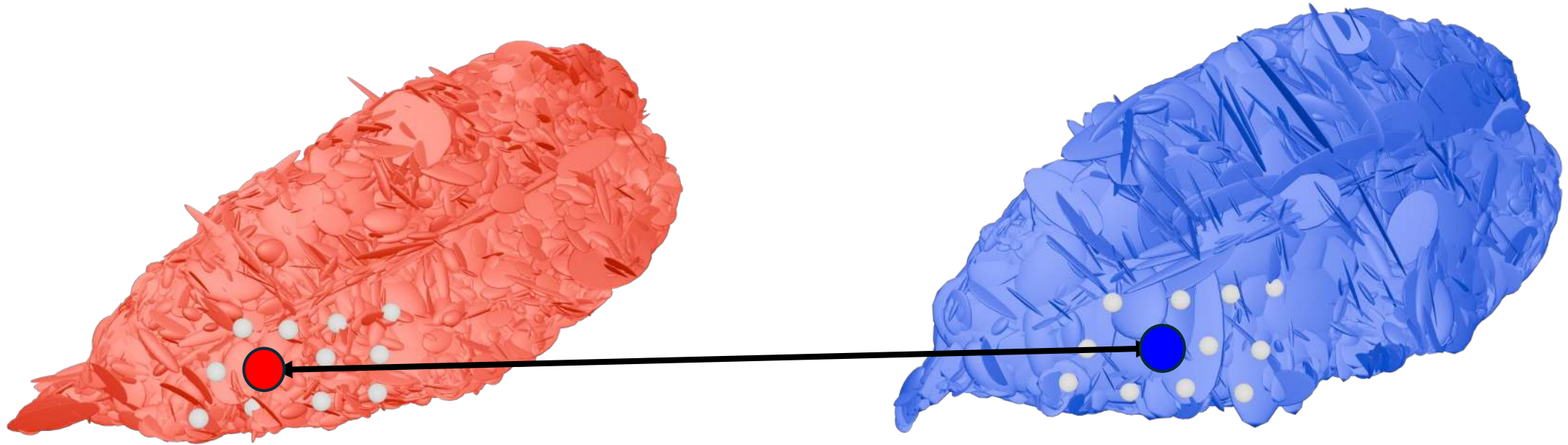
$$\sum_i w_i |T_i(p_i) - q_i|^2$$

Kernel Function Local Rigid Transformation

$$\frac{1}{|p_i - v|^{2\alpha}} \quad T(x) = Rx + t$$

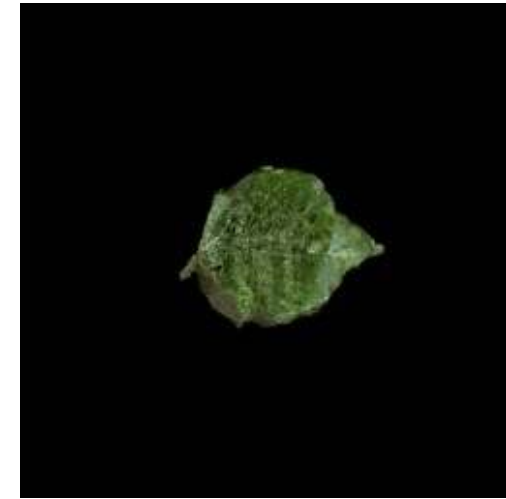
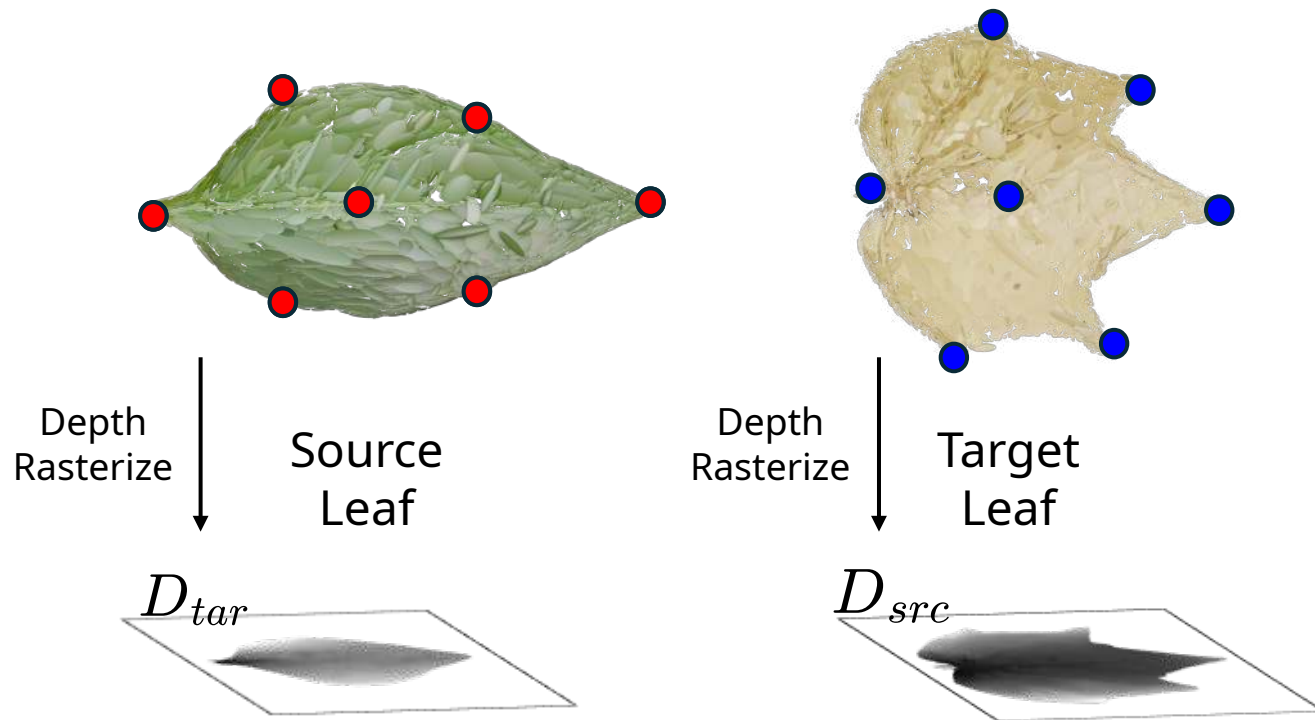
Initial Correspondence

- Align two leaves by PCA, build up initial correspondence.
- Farthest Point Sampling is applied to ensure uniform sampling.



Optimization

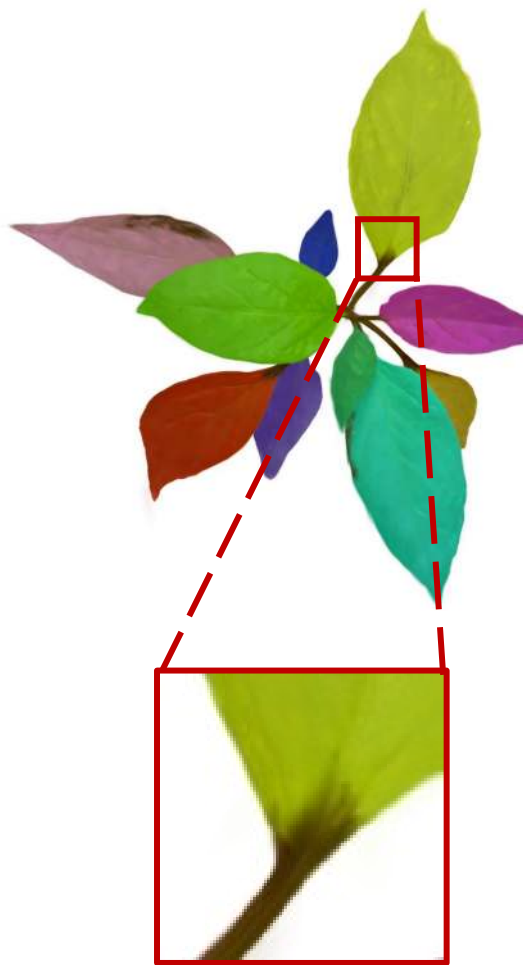
- Optimize correspondences with differentiable renderer.



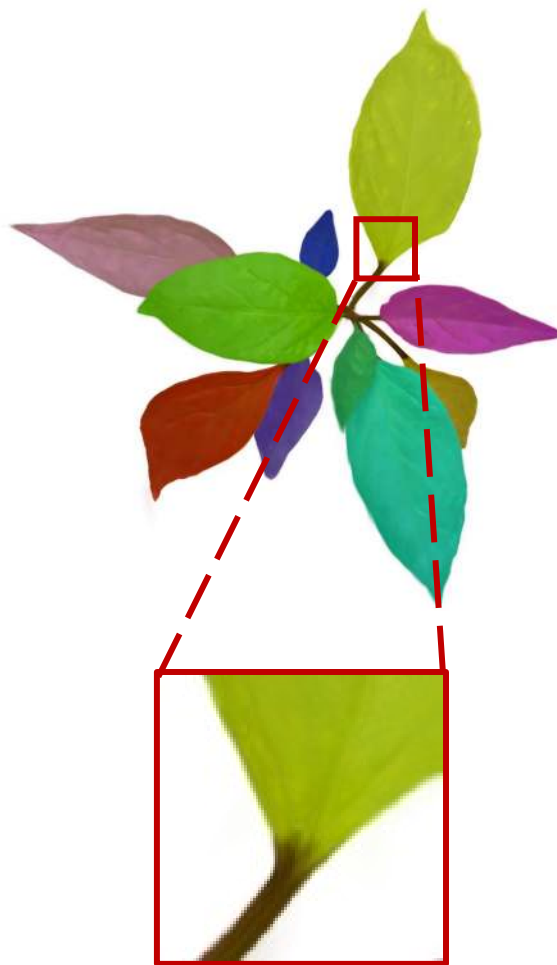
$$\mathcal{L}_{depth} = |D_{src} - D_{tar}|^2$$

$$\mathcal{L} = \mathcal{L}_{depth} + \lambda \mathcal{L}_{chamfer}$$

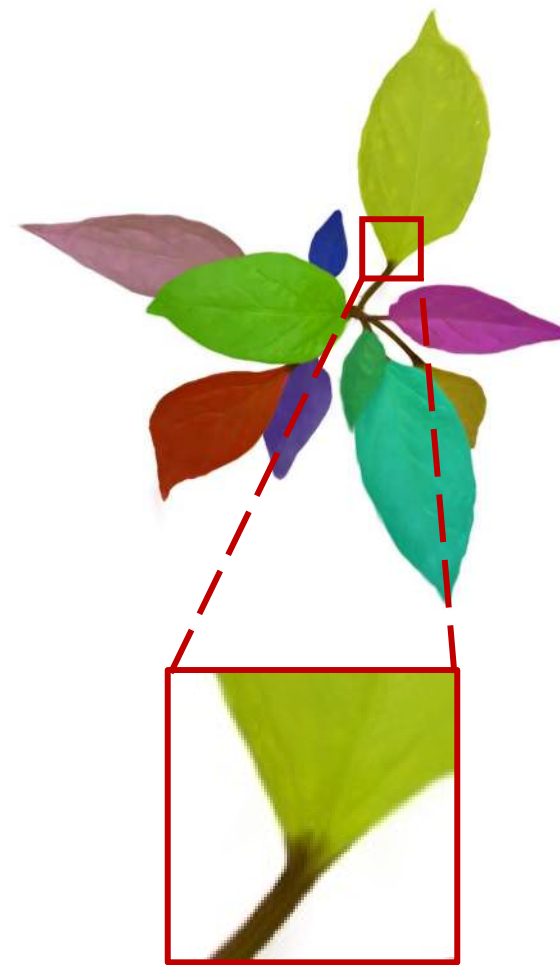
Segmentation comparison



Point
Transformer

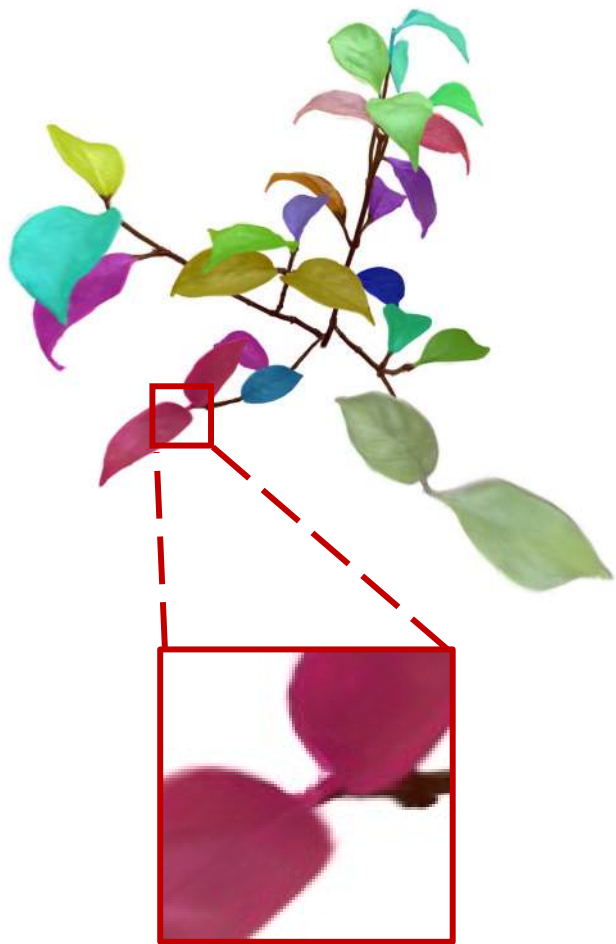


Our
s

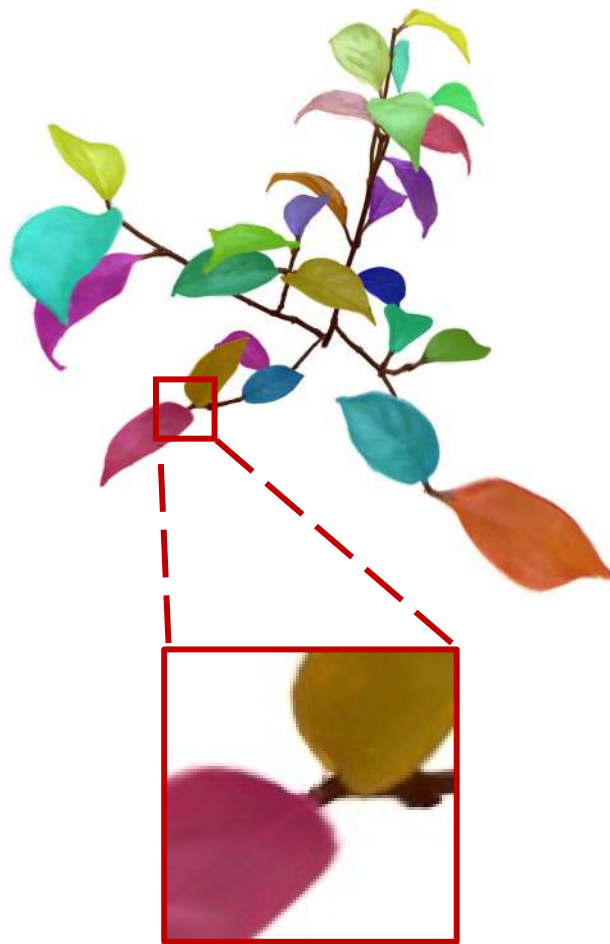


Ground
truth

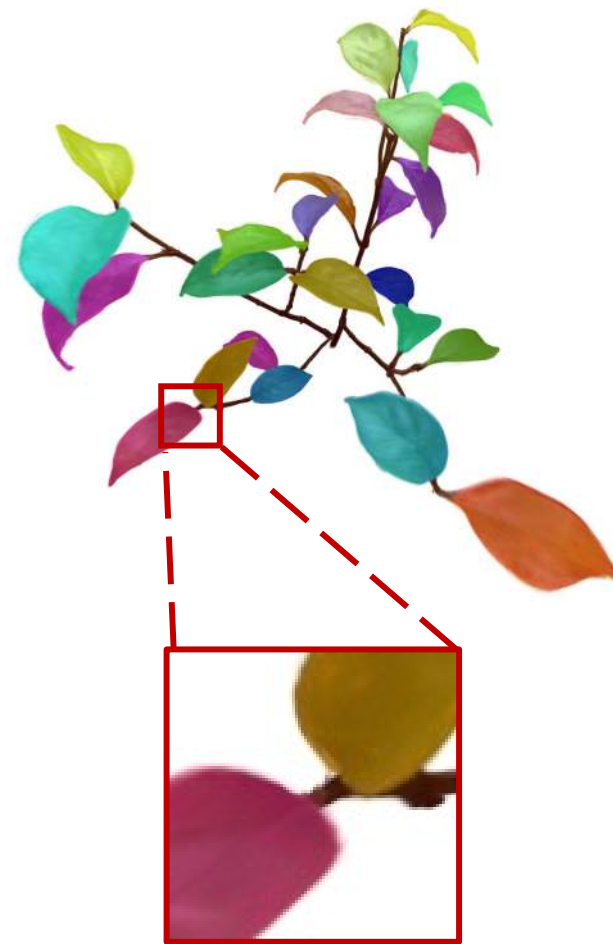
Segmentation comparison



Point
Transformer



Our
s



Ground
truth

Mesh comparison

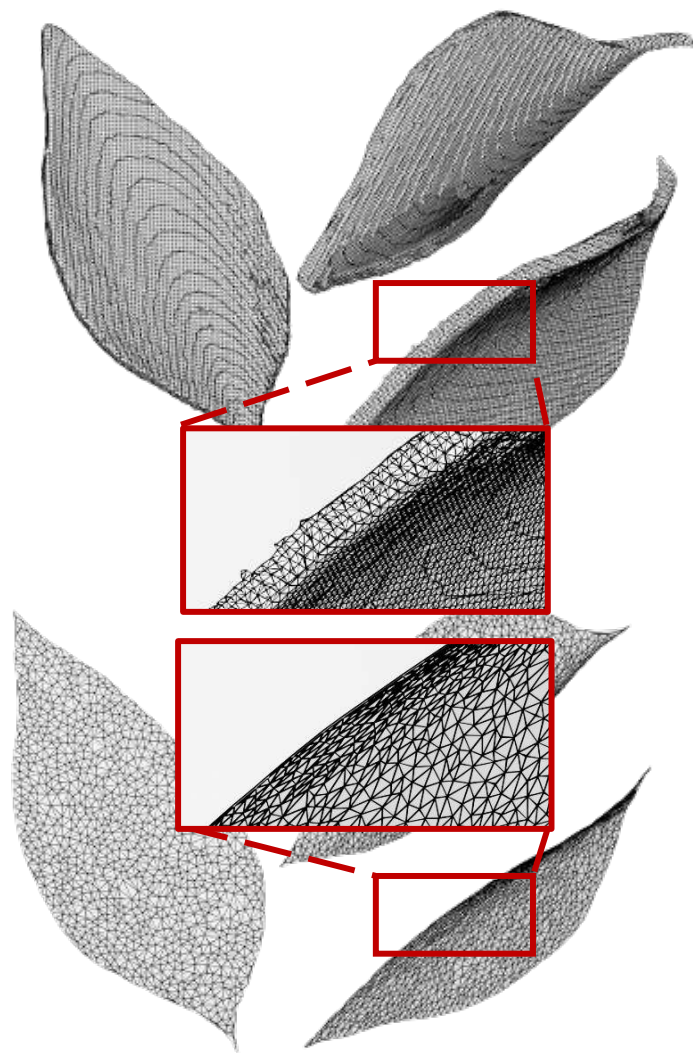
Input
Gaussian



GOF



Ours



Mesh comparison

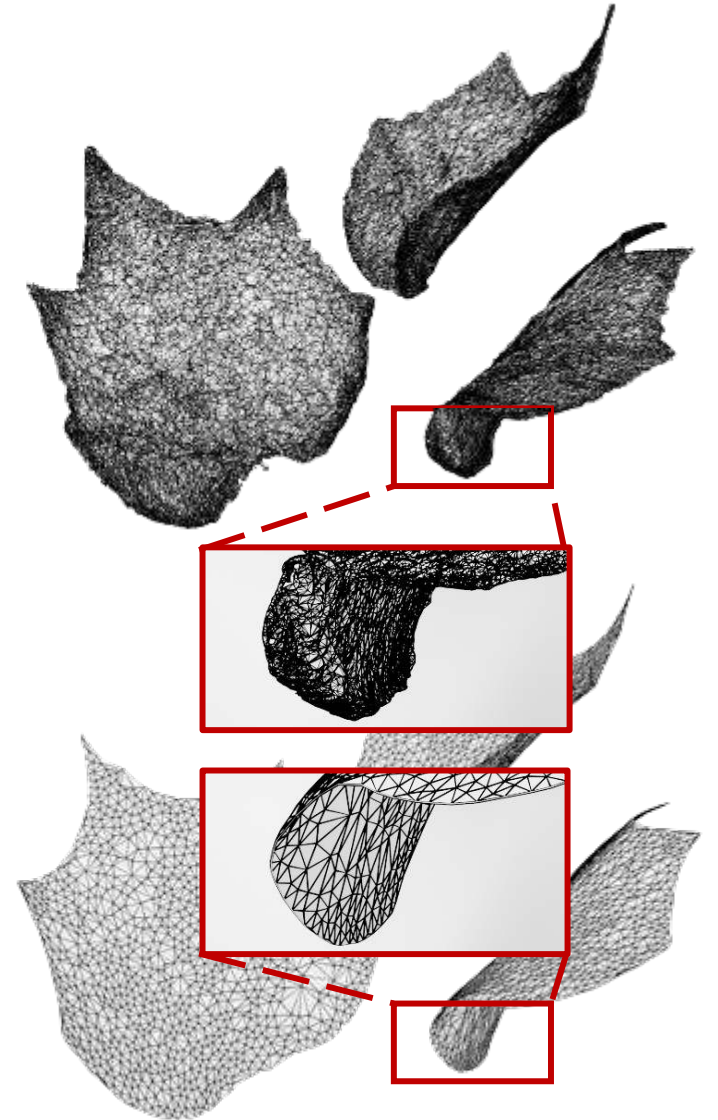
Input
Gaussian



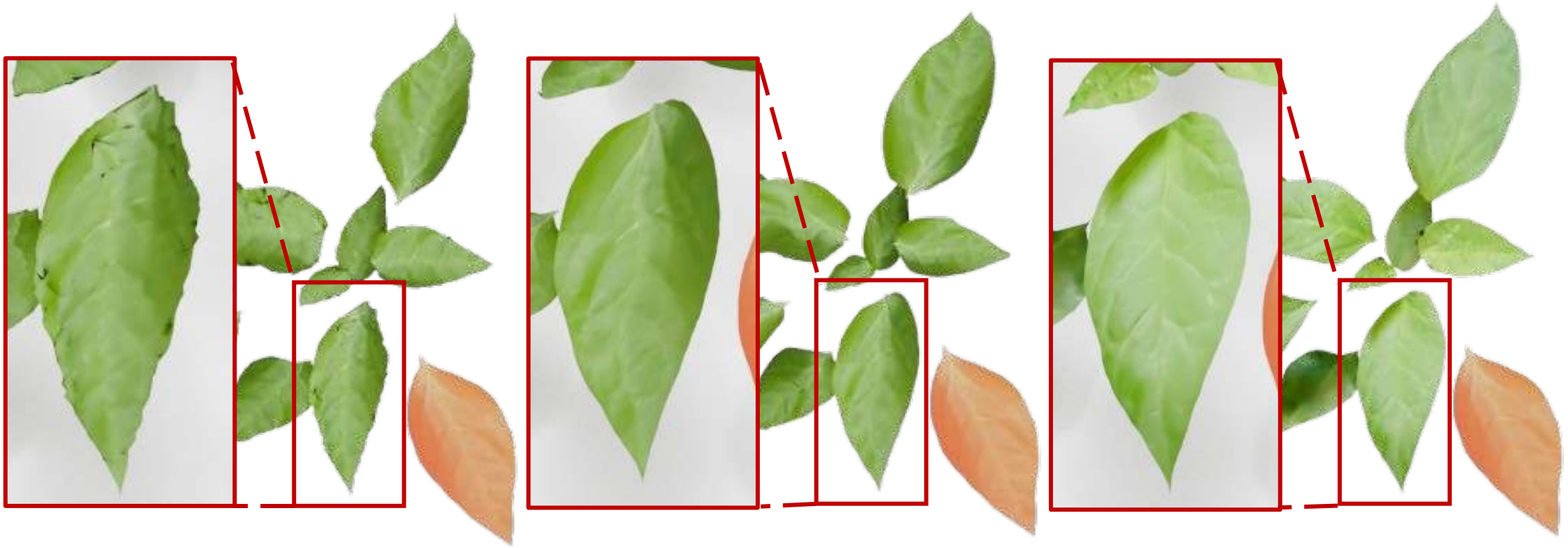
GOF



Ours



Deformation comparison



Non-Rigid ICP

Ours

Ground truth

Deformation comparison

PCA



NR-ICP



BCPD



Our
s



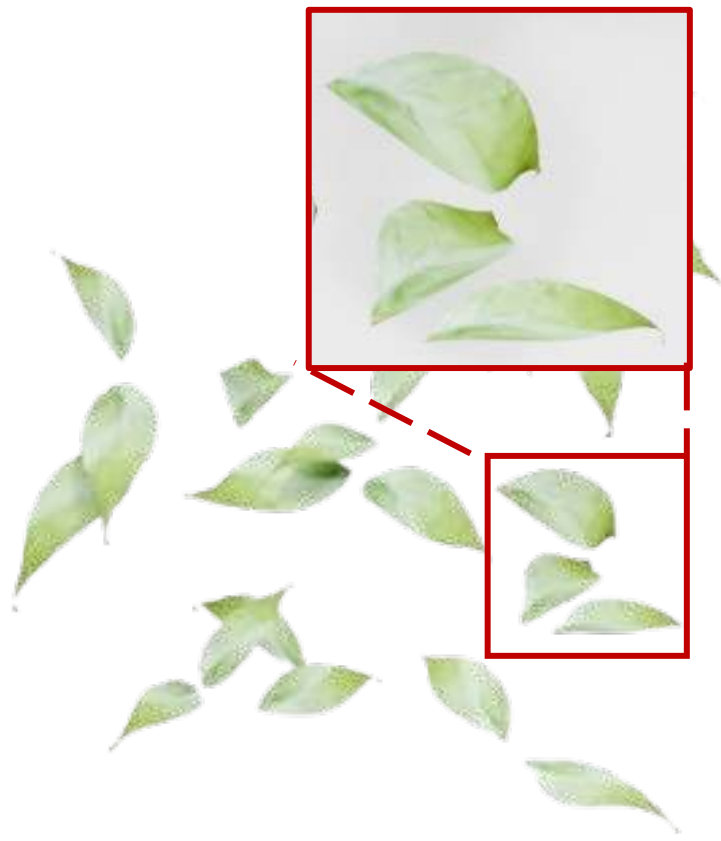
GT



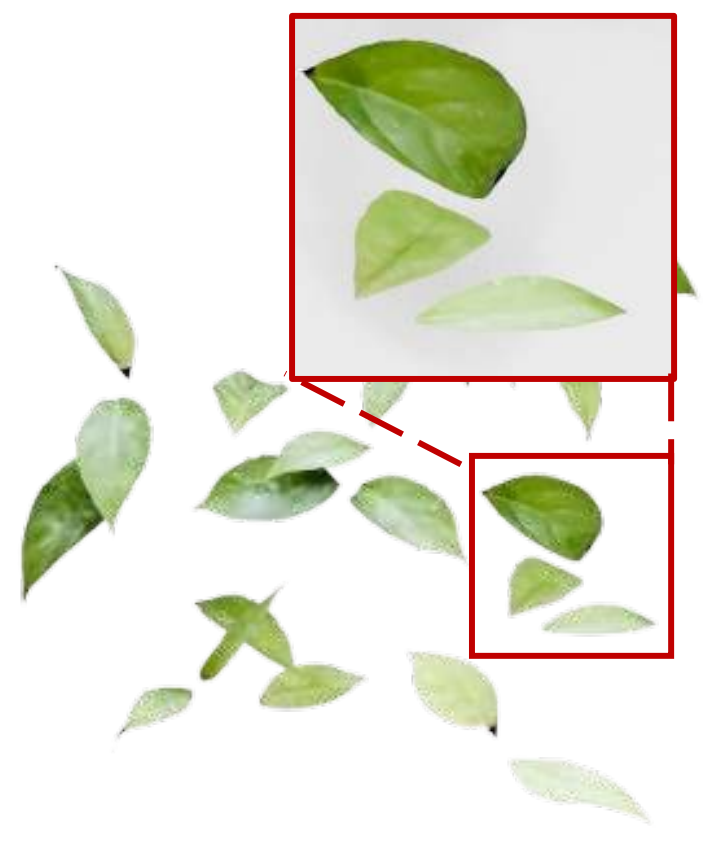
Deformation comparison



Non-Rigid ICP



Ours



Ground truth

Deformation comparison

PCA



NR-ICP



GT



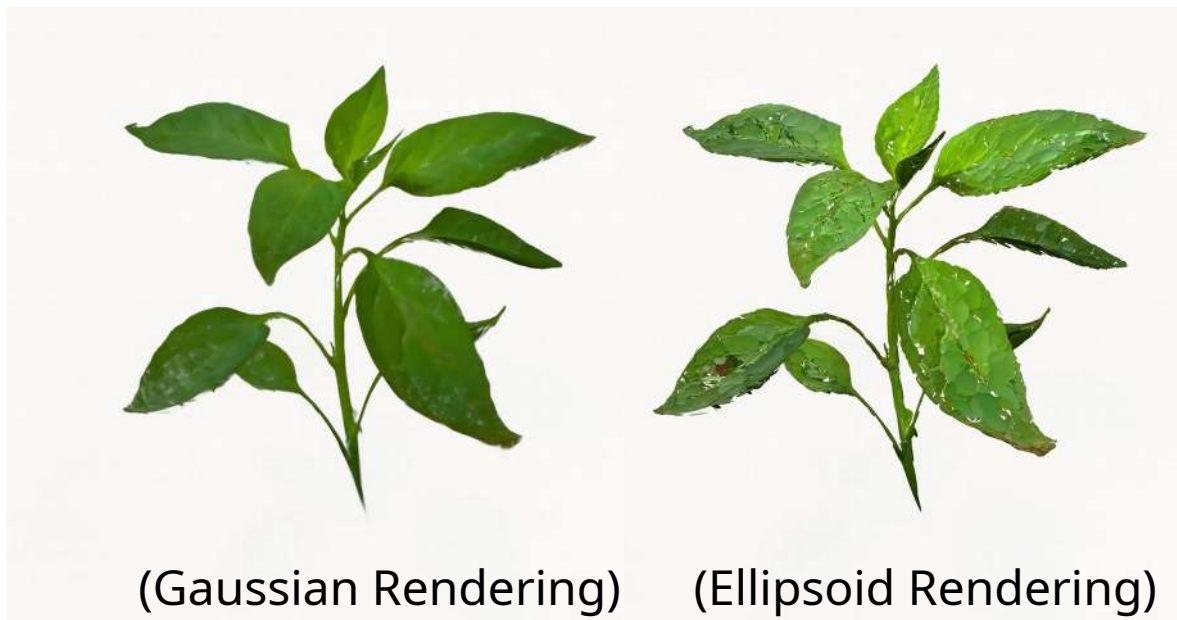
BCPD



**Our
s**



Results



Input 3DGS

Size: 11.9MB

#Primitive:

47k

GOF

Size: 26.2MB

#Primitive:

623k

Ours

Size: 0.75MB

#Primitive:

2k

Results



(Gaussian Rendering)



(Ellipsoid Rendering)



Input 3DGS

Size: 30.16MB

#Primitive:

122k

GOF

Size: 78.3MB

#Primitive:

1856k

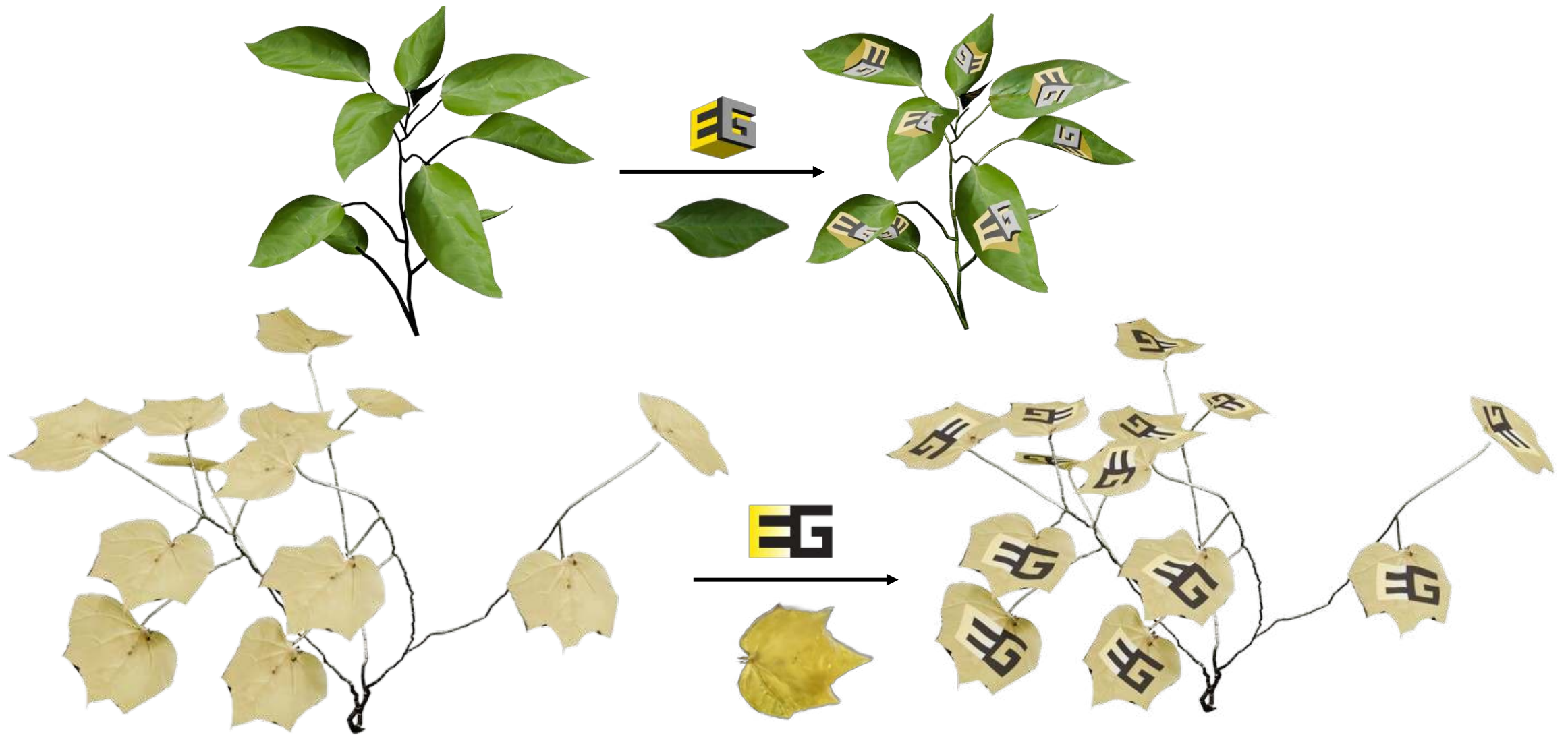
Ours

Size: 1.86MB

#Primitive:

2k

Texture In-painting



Texture In-painting

